Human adenovirus-36 antibody status is associated with obesity in children.

BACKGROUND: Human adenovirus-36 (Ad-36) is thought to induce obesity by a direct effect of the viral E4orf1 gene on lipogenic enzymes in host adipocytes. Ad-36 prevalence is 30% in obese adults, but prevalence has not been reported in childhood obesity. OBJECTIVES: To determine the prevalence of Ad-36 infection in obese Korean children (age 14.8 +/- 1.9; range 8.3-6.3 years); correlation of infection with BMI z-score and other obesity measures. METHODS: Blood was drawn at the annual school physical exam or clinic visit; Ad-36 status was determined by serum neutralization assay; and routine serum chemistry values. RESULTS: A total of 30% of subjects were positive (N = 25) for Ad-36; 70% were negative (N = 59). Significantly higher BMI z-scores (1.92 vs. 1.65, $p < 0.01$) and waist circumferences (96.3 vs. 90.7 cm, $p = 0.05$) were found in infected versus uninfected children. Cardiovascular risk factors were not significantly different. CONCLUSIONS: Ad-36 infection is common in obese Korean children and correlates highly with obesity. Ad-36 may have played a role in the obesity and Type 2 diabetes epidemic in children.